Cropping Plan and Rotation of a Family Farm

Introduction to Numerical Optimization – Project 2026

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A The coming year

A.1 The linear program

Indices: b block (B1–B4), c crop, t period (1 = spring, …, 4 = autumn). Units: ha, m^3, h, t, EUR.

A_b	block area (farm: 80 ha)	P_c	contract price	i_t	reservoir inflow
y_{bc}	yield per ha	F_c	free-market price	u_t	pump capacity
v_c	cost per ha	N_c, X_c	contract min, max	K, l_0, l_T	reservoir size, start, end
m_c	max farm share	g_{ct}, n_{ct}	water, hours per ha	L	yearly permit
σ	non-productive share	f_t, k_t, o_t	family hours, hire cap, wage	e	pumping cost per m^3

Variables (≥ 0): a_{bc} area of crop c on block b; s_c tonnes sold at the contract price; p_t pumped water; l_t reservoir level; w_t spilled water; h_t hired hours.

Shorthand: $T_c = \sum_b a_{bc}$ area of crop c; $Q_c = \sum_b y_{bc} a_{bc}$ harvest; $W_t = \sum_c g_{ct} T_c$ water used; $H_t = \sum_c n_{ct} T_c$ hours needed. If a crop has one price, $F_c = P_c$.

Objective:

$$\max\ G = \sum_c \left[P_c\, s_c + F_c\,(Q_c - s_c) \right] - \sum_c v_c\, T_c - \sum_t o_t\, h_t - e \sum_t p_t .$$

Constraints:

(1)	$\sum_c a_{bc} = A_b \ \ \forall b$	each block is fully sown (fallow = idle land)
(2)	$T_c \leq 80\, m_c$	no crop takes too much of the farm
(3)	$T_{\text{fallow}} \geq 80\, \sigma$	some land must rest
(4a)	$s_c \geq N_c$	contract: deliver the minimum
(4b)	$s_c \leq X_c$	the contract price is paid up to the ceiling
(4c)	$s_c \leq Q_c$	sell at most the harvest
(5)	$l_t = l_{t-1} + i_t + p_t - W_t - w_t$	reservoir balance
(6a)	$l_t \leq K$	reservoir size
(6b)	$p_t \leq u_t$	pump size
(7a)	$\sum_t p_t \leq L$	yearly permit
(7b)	$l_4 \geq l_T$	reserve at the end
(8a)	$H_t \leq f_t + h_t$	family hours plus hired hours
(8b)	$h_t \leq k_t$	limit on hired hours

A.2 Solution

Julia, JuMP and Gurobi (`src/a.jl`). The optimal gross margin is

$$G = \mathbf{122\,077.77}\ \text{EUR} = 248\,525.10\ \text{(revenue)} - 122\,547.32\ \text{(crops)} - 3\,000\ \text{(hired work)} - 900\ \text{(pumping)}.$$

crop (ha)	B1	B2	B3	B4	farm
wheat	0	18	0	14	32
barley	0	0.85	11.30	0	12.15
maize	0	0	3.50	0	3.50
sugar beet	11.11	3.15	0	0	14.26
potato	14.89	0	0	0	14.89
beans	0	0	0	0	0
fallow	0	0	3.20	0	3.20

Table 1: Optimal cropping plan, block by block.

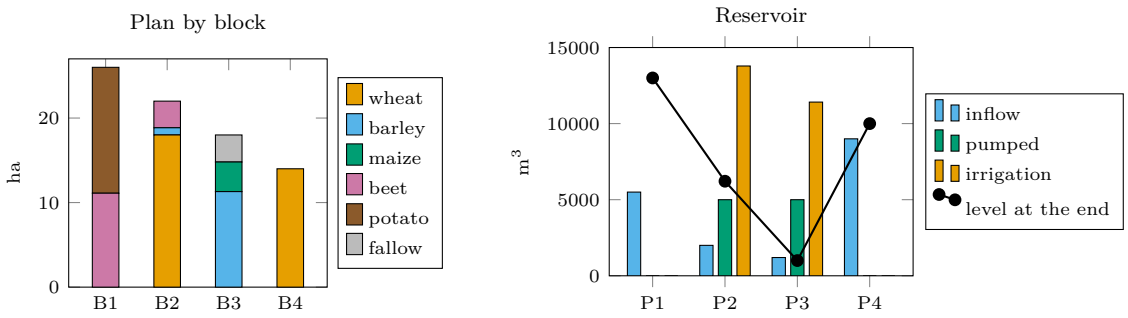


Figure 1: Left: the plan. Right: the reservoir is full after P1 (1 500 m^3 are lost), irrigation empties it in P2–P3, the autumn inflow refills it to the required 10 000.

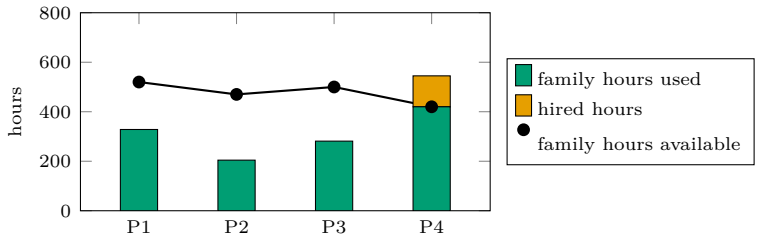


Figure 2: Labour. The family has spare hours in P1–P3. In autumn it is short, so the maximum of 125 h is hired.

- **Used up:** land, wheat share (32 ha), non-productive land (3.2 ha), potato contract ceiling (700 t), the permit (10 000 m^3), pumps in P2 and P3, reservoir size in P1, the final reserve, autumn labour (family hours and the maximum of hired hours).
- **Not used up:** the shares of all other crops, the sugar beet contract (299 t above the minimum, 501 t below the ceiling), family hours in P1–P3, pumps in P1 and P4.

B What the plan is worth

B.1 Standard form and dual

Standard form $\min\{c^\top x : \; Ax = b, \; x \geq 0\}$, with $z_k \geq 0$ added to each inequality:

$$
\begin{aligned}
\min \quad & \textstyle\sum_c v_c T_c - \sum_c \left[(P_c - F_c)\, s_c + F_c\, Q_c\right] + \sum_t o_t\, h_t + e \sum_t p_t \\
\text{s.t.} \quad (1) \quad & \textstyle\sum_c a_{bc} = A_b \\
(2) \quad & T_c + z_2 = 80\, m_c \\
(3) \quad & T_{\text{fallow}} - z_3 = 80\, \sigma \\
(4a) \quad & s_c - z_{4a} = N_c \\
(4b) \quad & s_c + z_{4b} = X_c \\
(4c) \quad & s_c - Q_c + z_{4c} = 0 \\
(5) \quad & l_t - l_{t-1} - p_t + W_t + w_t = i_t \\
(6a) \quad & l_t + z_{6a} = K \\
(6b) \quad & p_t + z_{6b} = u_t \\
(7a) \quad & \textstyle\sum_t p_t + z_{7a} = L \\
(7b) \quad & l_4 - z_{7b} = l_T \\
(8a) \quad & H_t - h_t + z_{8a} = f_t \\
(8b) \quad & h_t + z_{8b} = k_t \\
& a_{bc},\; s_c,\; p_t,\; l_t,\; w_t,\; h_t,\; z_k \;\geq\; 0
\end{aligned}
$$

Dual $\max\{b^\top\pi:\ A^\top\pi \le c\}$, one free variable π_k for each row (k):

$$\max \sum_b A_b\pi_1 + 80\sum_c m_c\pi_2 + 80\sigma\,\pi_3 + \sum_c N_c\pi_{4a} + \sum_c X_c\pi_{4b} + \sum_t i_t\pi_5 + l_0\pi_5 + K\sum_t \pi_{6a} + \sum_t u_t\pi_{6b} + L\pi_{7a} + l_T\pi_{7b} + \sum_t f_t\pi_{8a} + \sum_t k_t\pi_{8b}$$

$$\begin{aligned}
a_{bc}:&\ \ \pi_1 + \pi_2 + [c{=}\text{fallow}]\,\pi_3 - y_{bc}\,\pi_{4c} + \textstyle\sum_t \left(g_{ct}\,\pi_5 + n_{ct}\,\pi_{8a}\right) \le v_c - F_c\,y_{bc}\\
s_c:&\ \ \pi_{4a} + \pi_{4b} + \pi_{4c} \le F_c - P_c\\
p_t:&\ \ -\pi_5 + \pi_{6b} + \pi_{7a} \le e\\
l_t:&\ \ \pi_5(t) - \pi_5(t{+}1) + \pi_{6a} + [t{=}4]\,\pi_{7b} \le 0\\
w_t:&\ \ \pi_5 \le 0\\
h_t:&\ \ -\pi_{8a} + \pi_{8b} \le o_t\\
z_2, z_{4b}, z_{4c}, z_{6a}, z_{6b}, z_{7a}, z_{8a}, z_{8b}:&\ \ \pi_k \le 0\\
z_3, z_{4a}, z_{7b}:&\ \ -\pi_k \le 0
\end{aligned}$$

B.2 Meaning of the dual variables

We show each price as a gain of the margin when the right-hand side grows by one unit; this is $-\pi$ of B.1 (for every row;`src/print_b.jl`). Let $\alpha_b = -\pi_1$ be the price of land, $\beta_t = -\pi_5$ of water in the reservoir, $\gamma_t = -\pi_{8a}$ of a family hour.

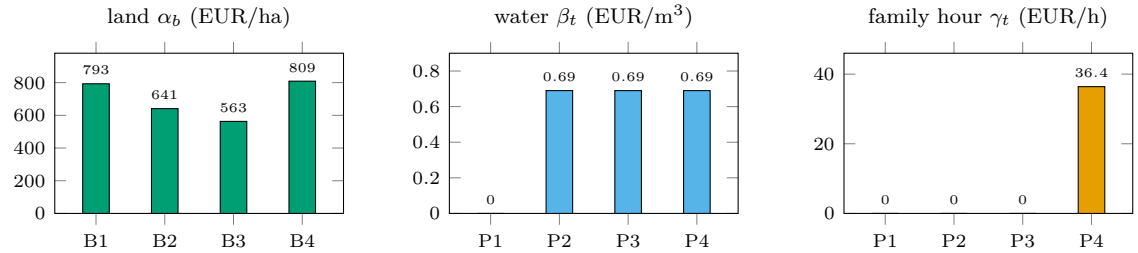


Figure 3: Dual variables of the land, reservoir and labour constraints.

- **Land** α_b (EUR/ha): one more hectare is worth 793, 641, 563, 809 EUR on B1–B4.
- **Water** β_t (EUR/m^3): 0 in P1, 0.69 in P2–P4. This is 0.60 (price of the permit) + 0.09 (pumping cost): an extra m^3 saves one pumped m^3.
- **Labour** γ_t (EUR/h): 0 in P1–P3, 36.40 in autumn. This is the wage 24 plus 12.40 for the hiring cap, which is reached.

Why is water worth 0 in P1? The reservoir is full and 1 500 m^3 overflow, so one more m^3 in spring is lost. From P2 on, stored water replaces pumped water, which is limited by the permit, so it has a value.

B.3 Primal and dual objective

The solver gives (for the maximization of G) primal 122 077.7731159276 and dual 122 077.7731159276. They agree up to 10^{-11}. In the form of B.1 ($\min -G$) both are $-122\,077.7731159276$.

B.4 The abstraction permit L

The current basis stays optimal for

$$\mathbf{9\,495.5} \ \le \mathbf{L} \le \mathbf{11\,000} \qquad (\Delta L \in [-504.5,\ +1000]).$$

Inside, the plan is a straight line in L. Each +1 000 m^3 moves these areas; all others stay fixed:

block	crop	change (ha per +1000 m^3)
B2	barley	+1.68
B2	sugar beet	−1.68
B3	barley	−4.02
B3	maize	+4.02
farm	barley / maize / beet	−2.34 / + 4.02 / − 1.68

The extra water irrigates maize on B3 (500 m^3/ha) instead of rain-fed barley. Maize needs 1.8 h/ha more than barley in autumn, where labour is exhausted, so beet on B2 (7.5 h/ha in autumn) turns into barley (3.2 h/ha): $4.02 \cdot 1.8 = 1.68 \cdot 4.3 = 7.2$ h. Water closes too: $4.02 \cdot 500 - 1.68 \cdot 600 = 1\,000$ m^3. The margin grows by 0.60 EUR per m^3, the price of the permit.

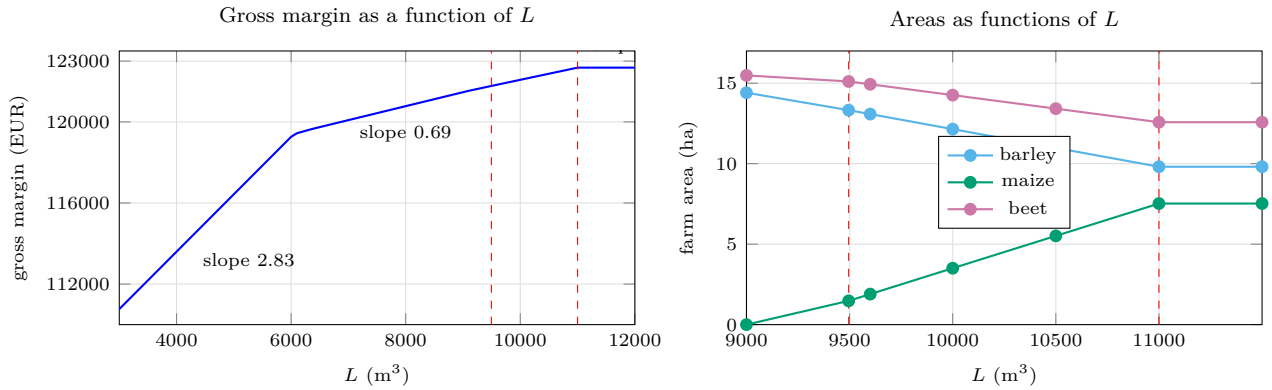


Figure 4: Left: margin against L (red: ends of the interval). Right: inside the interval the areas are straight lines.

- **Above 11 000:** nothing changes. The margin is flat at 122 677.73 EUR (slope 0).
- **Below 9 495.5:** the basis changes and the margin falls faster. The slope is 0.61, then 0.69 (down to $L \approx 6\,500$), then rises to 2.83 EUR/m^3 (below $L \approx 6\,000$).

The margin is concave in L: water goes to its best use first. The price 0.60 is valid only inside the interval.

B.5 Primal and dual simplex

	objective (EUR)	iterations	time (s)
primal simplex	122 077.77	29	< 0.002
dual simplex	122 077.77	39	< 0.002
+10 h autumn, primal (warm start)	122 391.96	3	< 0.002
+10 h autumn, dual (warm start)	122 391.96	2	< 0.002

- **Comparison.** Same optimum. 29 against 39 iterations, and the times are about 0. With 35 rows and 50 columns this proves nothing about which method is better: the counts depend on the start basis and on the solver's choices, and every run takes milliseconds.
- **+10 family hours in autumn** changes only the right-hand side b. The old basis stays dual feasible and may lose primal feasibility. The **dual simplex** is made for this case: it starts from the old basis and repairs it. Both methods need only 2–3 iterations from the old basis, against 29–39 from scratch.
- The margin grows by 314.18 EUR (31.4 EUR/h), less than the price 36.4, because the basis changes after some hours.

C The rotation

C.1 Three years with the rules

Variables get an upper index $y = 1, 2, 3$ (a^y_{bc}, G^y). Constraints (1)–(8) hold in every year. The reservoir carries over, $l^y_0 = l^{y-1}_4$; the reserve $l^3_4 \geq l_T$ is required only after year 3. The history a^0_{bc} is data. With discount rate $\rho = 0.03$:

$$\max \sum_{y=1}^{3} \frac{G^y}{(1+\rho)^{y-1}}.$$

(9)	$a^{y-1}_{bc} + a^y_{bc} \leq r_c\, A_b \quad \forall b, c$	return period (r_c = return share)
(10)	$a^y_{b,\text{potato}} \leq \sum_{c \in \text{cereals}} a^{y-1}_{bc} \quad \forall b$	potato only after cereals
(11)	$\sum_{c \in \text{legumes}} T^y_c \geq 80\,\lambda$	legumes ($\lambda = 0.1$: 8 ha of beans)

Year 0 appears only in (9) and (10) for $y = 1$, as numbers. Code: `src/c.jl`. The discounted gross margin is **339 933.66 EUR**.

crop (ha)	year 1 B1	B2	B3	B4	year 2 B1	B2	B3	B4	year 3 B1	B2	B3	B4
wheat	8.4	9.8	5.2	8.6	13.7	6.6	7.7	4.0	8.4	13.2	1.8	8.6
barley	0	0	0.9	4.8	0	0	0	5.0	0	0	2.2	4.8
maize	0	0	0.2	0	0	0	5.1	0	0	0	2.2	0
beet	10.9	4.8	0	0	6.0	5.3	0	0	10.9	1.4	0	0
potato	6.7	0	8.6	0	6.3	8.7	0.4	0	6.7	0	8.6	0
beans	0	7.4	0	0.6	0	1.4	1.6	5.0	0	7.4	0	0.6
fallow	0	0	3.2	0	0	0	3.2	0	0	0	3.2	0

Table 2: Cropping plans of years 1, 2, 3.

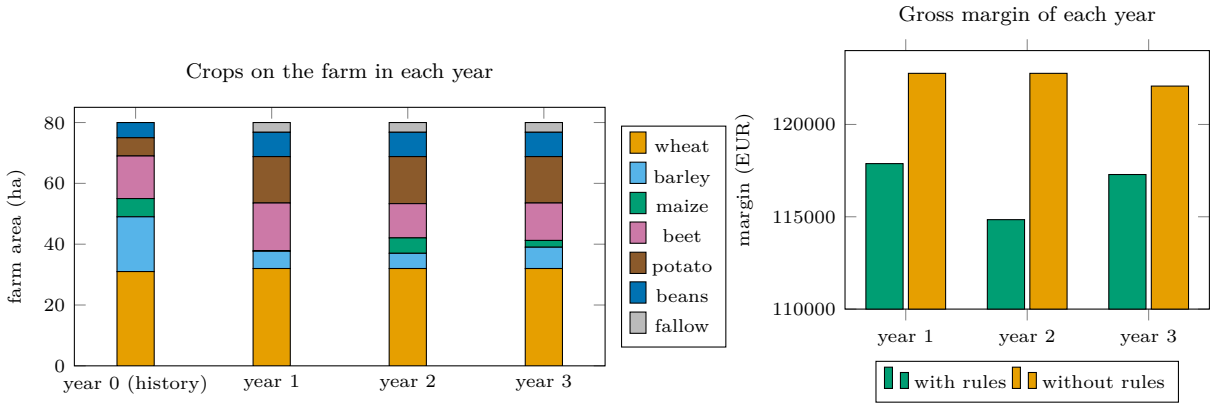


Figure 5: Left: crops on the whole farm; the history (year 0) grows 14 ha of beet and 6 ha of potato, the plan switches to potato and beans. Right: yearly margins (117 879.15, 114 843.90, 117 288.42 EUR; weights 1, 0.9709, 0.9426).

C.2 What the rotation costs

Without the three rules the discounted margin is 357 029.70 EUR. **The rules cost** 17 096 **EUR, i.e.** 4.79%**.**



Figure 6: Cost of the rules, in each case against the plan without any rule (357 029.70 EUR).

case	discounted margin (EUR)	cost (EUR)	cost (%)
all three rules	339 933.66	17 096.04	4.79
without the return period	346 411.47	10 618.24	2.97
without the preceding crop	339 933.66	17 096.04	4.79
without the legume rule	347 922.94	9 106.76	2.55
only return period	347 922.94	9 106.76	2.55
only preceding crop	355 549.84	1 479.86	0.41
only legume	347 318.64	9 711.06	2.72
no rule	357 029.70	0	0

Table 3: Discounted margin in each case; the cost is the difference to “no rule”.

Which rule is expensive? Remove one rule at a time:

- without legumes the farm gains 7 989 EUR;
- without the return period it gains 6 478 EUR;
- without the preceding-crop rule it gains nothing.

The legume rule is the expensive one. It costs more by both measures: 7 989 against 6 478 EUR when removed alone, and 9 711 against 9 107 EUR when kept alone.

Why the plan resists:

- **Return period.** The free plan repeats each crop on its best soil every year: 14 ha of wheat on B4 (heavy clay), 14.9 ha of potato and 11.1 ha of beet on B1 (deep loam). The rule caps a crop over two consecutive years at $r_c A_b$, e.g. 12.6 ha of wheat on B4 and 13 ha of potato on B1, so only about half of these areas can stay on their best block each year. The rest moves to blocks with lower yields: potatoes to B2 and B3, barley and beans onto B4.
- **Legume.** The free plan grows no beans: they have the lowest margin on every soil (351–554 EUR/ha, against at least 661 EUR/ha for any other productive crop). The rule forces 8 ha of beans every year in place of more profitable crops.
- **Preceding crop.** With the other two rules it is not binding: the return period already moves potatoes from B1 to B2 and B3, which carried wheat the year before. Alone it costs 1 480 EUR: the free plan grows potatoes on B1 every year with no cereal before them, so the farm must put wheat on B1 in some years to keep its potatoes there.

C.3 Three investments

Method. For each resource the program buys $1, 2, 3, \ldots$ units, solves the three years again and records the gain of the discounted margin. Gain per unit and per year (the discounted gain divided by $1+1.03^{-1}+1.03^{-2} = 2.9135$) is the *value of a unit*. An investment pays while its price is below this value (Figure 7).

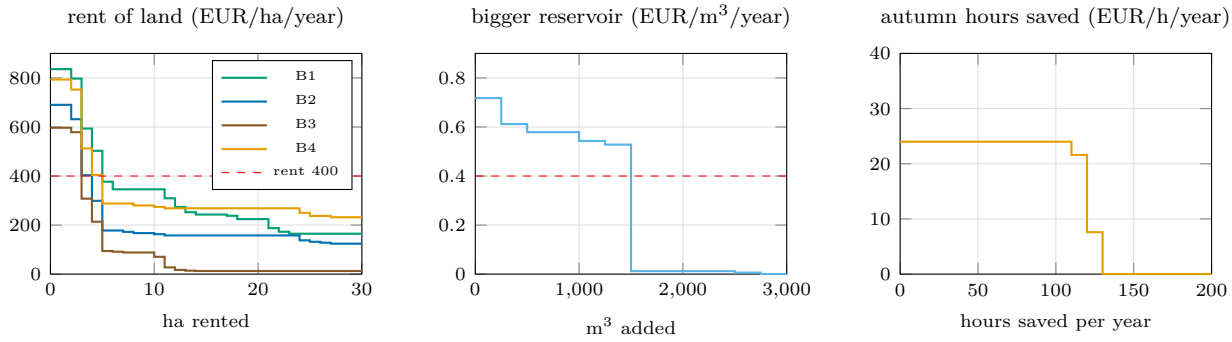


Figure 7: Value of one more unit per year (solid) and the assumed price per year (dashed): rent 400 EUR/ha, reservoir annuity 0.40 EUR/m^3.

The first units are the most valuable; then the value falls. Ranking by the first unit: land B1 (836 EUR/ha) > B4 (794) > B2 (690) > B3 (597); autumn hours saved (24 EUR/h); reservoir (0.72 EUR/m^3); permit (0.21 EUR/m^3).

Assumptions (not in the data):

- land: rent 400 EUR/ha per year, 3-year contract; the parcel lies next to B1 and has the same soil (deep loam); the 5 ha were under cover fallow in year 0; the shares of the farm (diversification, fallow, legume) are computed from 85 ha;
- reservoir: 6 EUR per m^3, life 20 years, paid as an annuity at 3%: 604.94 EUR per year for 1 500 m^3;
- machine (for example a harvester) that saves 100 autumn hours per year: price 30 000 EUR, life 10 years, paid as an annuity at 3%: 3 516.92 EUR per year.

investment	parameter changed	gain (EUR)	cost (EUR)	net (EUR)	decision
rent 5 ha on B1	area of B1: 26 → 31 ha (the farm: 80 → 85) in all three years	10 383.71	5 826.94	+4 556.77	**yes**
reservoir +1 500 m^3	capacity K: 13 000 → 14 500	2 584.83	1 762.48	+822.35	**yes**
machine saving 100 autumn hours per year	autumn hours f_4: 420 → 520	6 992.69	10 246.43	−3 253.74	no

Table 4: The three investments (discounted over the three years).

- **Rent B1: yes.** 5 ha is the amount where the next hectare (377) is worth less than the rent (400).
- **Bigger reservoir: yes**, up to 1 500 m^3.
- **Machine: no.** A harvester costs about 30 000 EUR (10 years, 3%), i.e. 10 246.43 EUR over the three years, more than the 6 992.69 EUR it saves. It would pay only below 20 470 EUR.

The decisions depend on the assumed prices; the curves show the answer for any other price.